

KUGAARUK, NU, Canada

WMO: 714070

Lat: 68.538N

Lon: 89.789W

Elev: 52

StdP: 14.67

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-44.8	-42.0	-52.3	0.2	-44.0	-49.7	0.3	-41.6	32.8	-17.5	29.6	-20.2	3.9	60	1.062

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	66.9	55.4	63.0	53.4	58.9	51.4	56.3	64.8	53.8	61.5	51.6	58.7	10.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.8	57.2	59.8	49.1	51.8	57.7	46.7	47.2	54.9	24.0	65.0	22.5	61.4	21.2	59.0	64.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.3	22.7	-49.8	73.2	4.1	3.9	-52.7	76.0	-55.2	78.2	-57.5	80.4	-60.5	83.2
			-48.9	61.0	3.1	2.8	-51.1	63.0	-52.9	64.7	-54.7	66.2	-56.9	68.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-23.3	-25.3	-19.3	-2.4	18.0	37.9	49.2	45.7	33.5	18.4	-3.4	-16.3		
	DBStd	28.30	10.55	11.11	11.14	11.10	9.51	7.08	6.15	6.21	5.80	8.90	12.33	12.33		
	HDD50	14856	2272	2108	2148	1573	992	371	91	166	495	981	1603	2056		
	HDD65	20225	2737	2528	2613	2023	1457	812	492	598	945	1446	2053	2521		
	CDD50	107	0	0	0	0	0	9	65	33	0	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	5	0	0	0	0	0	0	5	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	11.0	11.7	11.3	11.6	11.1	10.6	10.2	9.9	10.1	10.7	11.4	11.8	11.5		
	PrecAvg	8.9	0.3	0.2	0.5	0.6	0.5	0.9	1.5	1.7	1.1	0.8	0.4	0.4		
	PrecMax	15.9	0.7	0.5	2.4	3.3	2.3	3.3	5.1	4.8	2.4	1.6	0.9	0.9		
(15) Precipitation	PrecMin	4.8	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.3	0.3	0.1	0.0		
	PrecStd	2.5	0.2	0.1	0.6	0.7	0.5	0.6	1.1	1.0	0.5	0.3	0.2	0.2		
	DB	11.5	9.5	15.0	29.8	40.1	68.2	73.3	70.0	54.4	36.5	27.6	23.0			
	MCWB	10.6	8.8	13.7	28.7	35.8	55.2	59.2	57.4	46.1	34.6	26.7	22.1			
	2% DB	4.9	4.6	9.4	23.4	35.5	60.6	67.9	65.2	49.3	33.9	23.9	14.8			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	4.5	3.8	8.4	22.0	33.3	51.6	55.7	55.2	44.1	32.4	22.9	13.9			
	5% DB	-1.7	-1.1	4.0	19.9	33.3	53.8	64.8	61.1	45.3	31.8	19.5	8.1			
	MCWB	-2.4	-1.8	3.2	18.6	31.6	46.8	54.3	53.2	41.9	30.4	18.5	7.4			
	10% DB	-7.1	-8.4	-0.8	15.9	30.9	48.6	60.9	56.7	42.0	29.7	15.0	2.2			
	MCWB	-7.6	-8.9	-1.4	14.8	29.2	43.5	52.0	50.6	39.5	28.5	14.1	1.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	10.9	8.8	13.8	28.5	36.6	55.3	61.2	59.0	47.1	35.0	27.0	22.3			
	MCDB	11.4	9.2	14.7	30.1	39.7	67.2	70.2	67.6	51.9	36.4	27.7	23.0			
	2% WB	4.3	3.8	8.5	22.0	33.4	51.4	57.3	55.9	44.4	32.6	22.8	13.9			
	MCDB	4.7	4.5	9.4	23.2	35.0	60.2	66.1	63.5	48.8	33.7	23.7	14.7			
	5% WB	-2.3	-1.6	3.3	18.7	31.8	47.1	54.7	53.6	42.0	30.6	18.7	7.4			
(33) Mean Daily Temperature Range	MCDB	-1.8	-0.9	3.9	19.8	33.4	53.3	62.8	59.9	45.2	31.7	19.6	8.0			
	10% WB	-7.5	-8.8	-1.5	14.9	29.3	43.6	52.4	51.1	39.5	28.5	14.2	1.5			
	MCDB	-7.0	-8.5	-0.8	16.0	30.8	48.3	60.3	56.7	42.1	29.6	14.9	2.1			
	5% DB	11.7	11.7	13.6	15.3	10.9	10.4	14.5	12.7	7.6	8.9	11.3	11.3			
	MCDBR	16.9	17.0	19.2	15.5	9.6	20.0	21.2	20.5	14.6	9.0	14.0	15.3			
(36) Clear-Sky Solar Irradiance	MCWBR	16.7	16.7	18.8	14.9	8.4	13.0	11.8	12.2	10.6	8.8	13.9	15.1			
	5% WB	16.7	16.9	19.3	15.3	9.5	19.4	19.6	18.9	14.1	8.8	13.5	14.9			
	MCWBR	16.6	16.6	18.9	14.8	8.5	12.8	12.0	12.1	10.7	8.7	13.4	14.7			
	taub	0.149	0.206	0.255	0.301	0.327	0.323	0.357	0.325	0.295	0.239	0.156	N/A			
	taud	1.886	2.074	2.167	2.095	2.104	2.337	2.377	2.489	2.450	2.088	1.904	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	36	193	249	265	270	278	262	261	238	164	27	0			
	Edh at Noon	2	14	24	36	40	33	31	25	19	13	2	0			
(44) All-Sky Solar Radiation	RadAvg	8	140	635	1400	1949	1915	1636	1102	535	168	26	0			
	RadStd	1	10	23	46	112	162	159	86	45	18	2	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+1.64	+1.71	N/A	N/A	N/A	N/A	-785	-778	N/A	N/A

Nomenclature: See separate page